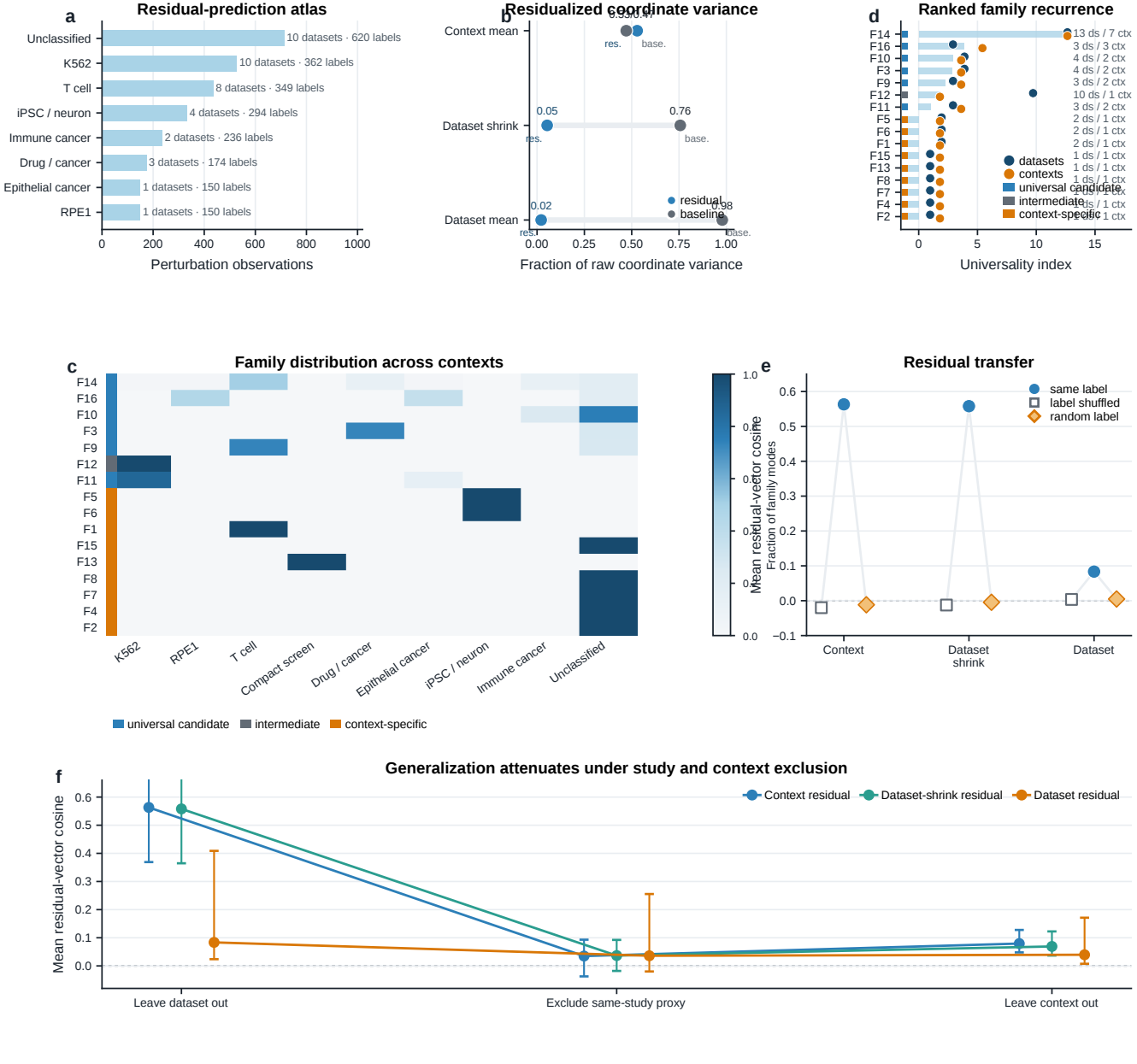
a
Residual-prediction atlas
Unclassified
K562
T cell
iPSC / neuron
Immune cancer
Drug / cancer
Epithelial cancer
RPE1
10 datasets · 620 labels
10 datasets · 362 labels
8 datasets · 349 labels
4 datasets · 294 labels
2 datasets · 236 labels
3 datasets · 174 labels
1 datasets · 150 labels
1 datasets · 150 labels
0
200
400
600
800
1000
Perturbation observations
b
Residualized coordinate variance
Context mean
Dataset shrink
Dataset mean
0.93/0.47
res.
base.
0.05
0.76
0.02
0.98
residual
baseline
0.00
0.25
0.50
0.75
1.00
Fraction of raw coordinate variance
d
Ranked family recurrence
F14
F16
F10
F3
F9
F12
F11
F5
F6
F1
F15
F13
F8
F7
F4
F2
13 ds / 7 ctx
3 ds / 3 ctx
4 ds / 2 ctx
4 ds / 2 ctx
3 ds / 2 ctx
10 ds / 1 ctx
3 ds / 2 ctx
2 ds / 1 ctx
2 ds / 1 ctx
2 ds / 1 ctx
1 ds / 1 ctx
1 ds / 1 ctx
1 ds / 1 ctx
1 ds / 1 ctx
1 ds / 1 ctx
1 ds / 1 ctx
datasets
contexts
universal candidate
intermediate
context-specific
0
5
10
15
Universality index
c
Family distribution across contexts
F14
F16
F10
F3
F9
F12
F11
F5
F6
F1
F15
F13
F8
F7
F4
F2
K562
RPE1
T cell
Compact screen
Drug / cancer
Epithelial cancer
iPSC / neuron
Immune cancer
Unclassified
universal candidate
intermediate
context-specific
1.0
0.0
Mean residual-vector cosine
e
Residual transfer
same label
label shuffled
random label
Fraction of family modes
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
Context
Dataset shrink
Dataset
f
Generalization attenuates under study and context exclusion
Context residual
Dataset-shrink residual
Dataset residual
Mean residual-vector cosine
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Leave dataset out
Exclude same-study proxy
Leave context out